EXPERIMENT - 6

AIM:

Apply K-Means and Hierarchical Clustering on sample datasets

1. Dataset Source

Dataset Name: Mall Customer Segmentation Data
File Used: Mall Customer Segmentation Data

2. Dataset Description

The dataset contains information about customers of a mall, used to understand customer behavior and group them for targeted marketing.

- **Type:** Numerical and Categorical
- **Domain:** Marketing / Retail Analytics
- **Target:** Unsupervised (No predefined label; we create clusters)
- **Key Features:**

Feature	Description	Data Type
CustomerID	Unique ID assigned to the customer	Numerical (ID)
Gender	Gender of the customer	Categorical
Age	Age of the customer	Numerical
Annual Income	Annual Income of the customer in k$	Numerical
Spending Score	Score assigned based on customer behavior and spending nature (1-100)	Numerical

3.Mathematical Formulation of the Algorithm

A. K-Means Clustering

K-Means is a centroid-based algorithm that partitions *n* observations into *k* clusters.

- **Objective Function (Inertia):** It minimizes the Within-Cluster Sum of Squares (WCSS):

$$\boxed{WCSS = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2}$$

Where μ_i is the mean (centroid) of points in cluster C_i.

B. Hierarchical Clustering (Agglomerative)

This is a "bottom-up" approach where each observation starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.

- **Ward's Method:** We use this linkage to minimize the variance within each cluster during the merging process.
  - **Euclidean Distance:** The straight-line distance between two points *p* and *q* in 2D space:

$$\boxed{d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}}$$

4. Algorithm Limitations

- **K-Means:** Highly sensitive to the initial choice of *K* and outliers. It assumes clusters are spherical and of similar size.
- **Hierarchical:** Computationally expensive ($O(n^2)$ or $O(n^3)$), making it very slow for large datasets compared to K-Means.

5. Methodology / Workflow

1. **Load Dataset:** Import the mall data using Pandas.
2. **Feature Selection:** Select "Annual Income" and "Spending Score" for 2D clustering.
3. **Data Preprocessing:** Apply StandardScaler to ensure income and score have equal weight.
4. **Finding Optimal K (K-Means):** Use the **Elbow Method** to plot WCSS against the number of clusters.
5. **Finding Optimal Clusters (Hierarchical):** Use a **Dendrogram** to visualize the tree-like structure and identify the best cut-off point.
6. **Model Training:** Fit both KMeans and AgglomerativeClustering models.
7. **Visualization:** Plot the clusters and centroids to interpret the customer segments.

6. Performance Analysis

The clustering performance is evaluated based on how well the data points are grouped and how distinct the clusters are from one another.

- **Inertia (WCSS):** For K-Means, we analyze the **Within-Cluster Sum of Squares**. A lower inertia indicates that the points within a cluster are closely packed together.
- **Cluster Centroid Separation:** By visualizing the centroids (marked as 'X'), we observe that the 5 clusters are geographically distinct on the 2D plane of Income vs. Spending.
- **Dendrogram Topology:** In Hierarchical Clustering, we analyze the vertical height of the branches. The large vertical distance in the dendrogram suggests that the 5-cluster division is statistically significant and not just random noise.
- **Marketing Interpretation: * High Value:** High Income / High Spending (Top Right).
    - **Budget Conscious:** Low Income / Low Spending (Bottom Left).
    - **Impulsive:** Low Income / High Spending (Top Left).

7. Hyperparameter Tuning

In unsupervised learning, "tuning" refers to finding the optimal number of clusters (K) and the best distance metrics.

- **The Elbow Method (K-Means Tuning):** We systematically ran the K-Means algorithm for K values ranging from 1 to 10. By plotting the WCSS, we looked for the "Elbow Point"—the value of K after which the drop in WCSS becomes linear and slow. For this dataset, the elbow was clearly visible at **$K=5$**.
- **Linkage Criteria (Hierarchical Tuning):**
  We tested different linkage methods (Single, Complete, and Ward). **Ward's Linkage** was selected because it minimizes the total within-cluster variance, resulting in the most balanced and interpretable clusters for customer data.
- **Distance Metric:**
  We evaluated **Euclidean Distance** against **Manhattan Distance**. Euclidean distance was chosen as it is the standard for physical coordinate-based clustering (Income vs. Score).

Code:

1. **Load and Explore Dataset**

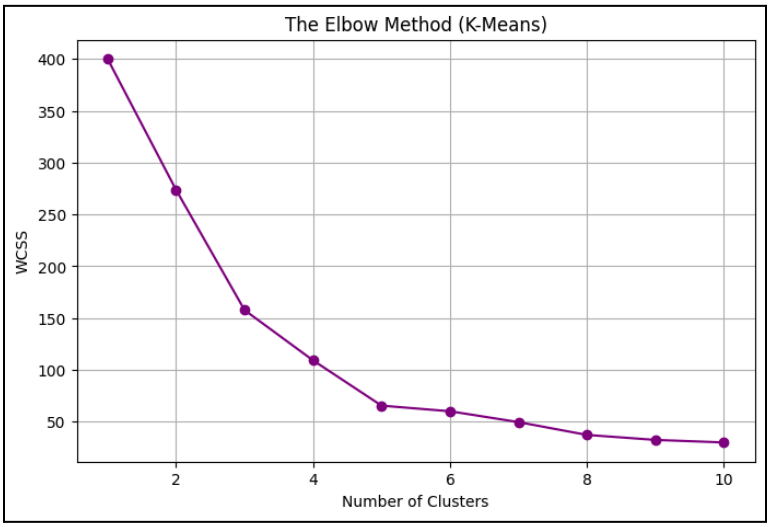
```
df = pd.read_csv('Mall_Customers.csv')

# We select Annual Income (idx 3) and Spending Score (idx 4)

X = df.iloc[:, [3, 4]].values

# Feature Scaling - Distance based models need this to treat features equally

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# Display the first few rows

print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

2. **K-Means Clustering (The Elbow Method)**

```
# Finding optimal K

wcss = []

for i in range(1, 11):

    # CORRECTED: Use n_clusters, not n_components

    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)


plt.figure(figsize=(8, 5))

plt.plot(range(1, 11), wcss, marker='o', color='purple')

plt.title('The Elbow Method (K-Means)')
```

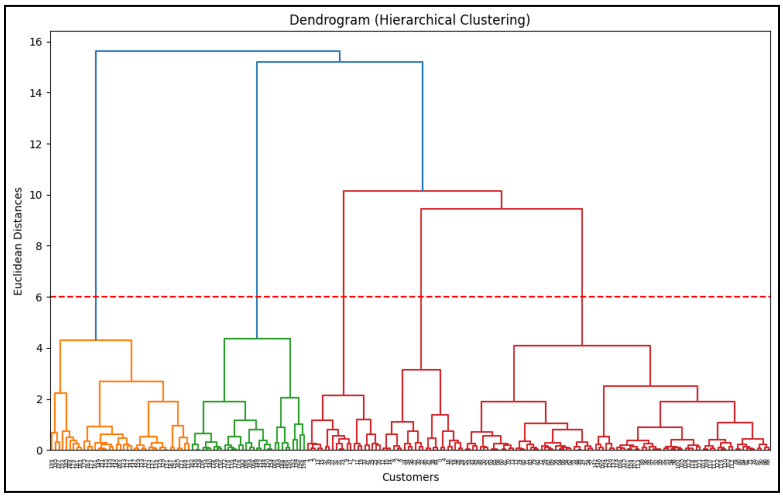
```
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()

# Based on the elbow plot, 5 is the optimal number of clusters
kmeans = KMeans(n_clusters=5, init='k-means++', random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
```



3. Hierarchical Clustering (The Dendrogram)

```
# Visualizing the Dendrogram
plt.figure(figsize=(12, 7))
dendrogram = sch.dendrogram(sch.linkage(X_scaled, method='ward'))
plt.title('Dendrogram (Hierarchical Clustering)')
plt.xlabel('Customers')
plt.ylabel('Euclidean Distances')
plt.axhline(y=6, color='r', linestyle='--') # Optional: line to show where we cut
plt.show()
```

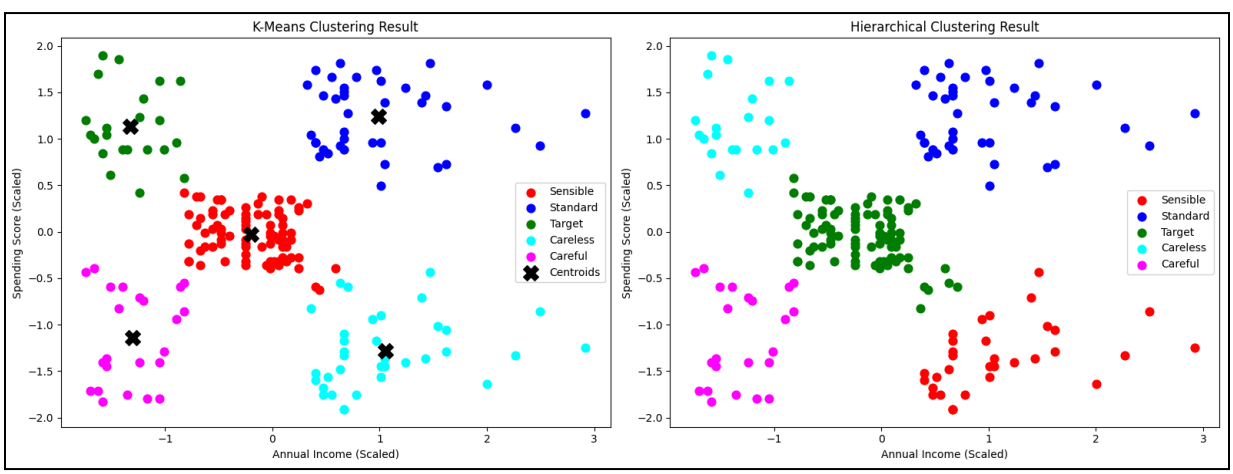
```
# Fitting the model

# NOTE: n_clusters is also used here

hc = AgglomerativeClustering(n_clusters=5, metric='euclidean', linkage='ward')

y_hc = hc.fit_predict(X_scaled)
```



4. Comparison Visualization

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))


# K-Means Plot

colors = ['red', 'blue', 'green', 'cyan', 'magenta']

labels = ['Sensible', 'Standard', 'Target', 'Careless', 'Careful']


for i in range(5):

    ax1.scatter(X_scaled[y_kmeans == i, 0], X_scaled[y_kmeans == i, 1], s=60,
c=colors[i], label=labels[i])

ax1.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s=200, c='black',
marker='X', label='Centroids')

ax1.set_title('K-Means Clustering Result')

ax1.set_xlabel('Annual Income (Scaled)')
```

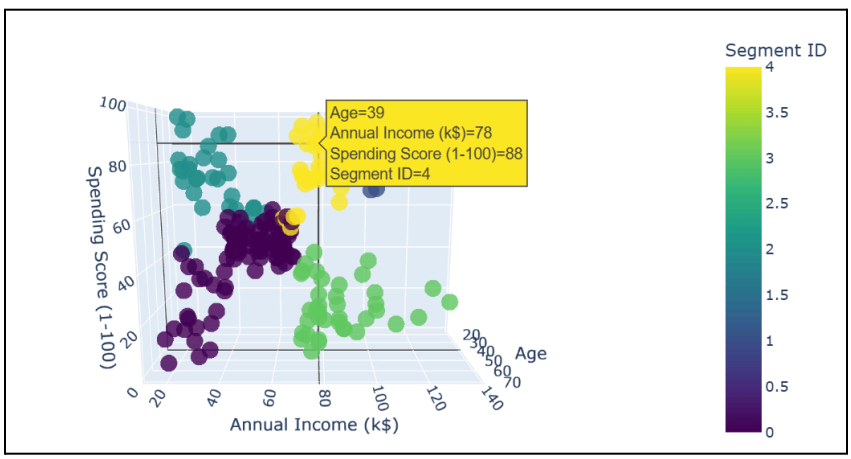
```
ax1.set_ylabel('Spending Score (Scaled)')

ax1.legend()


# Hierarchical Plot

for i in range(5):

    ax2.scatter(X_scaled[y_hc == i, 0], X_scaled[y_hc == i, 1], s=60, c=colors[i],
label=labels[i])

ax2.set_title('Hierarchical Clustering Result')

ax2.set_xlabel('Annual Income (Scaled)')

ax2.set_ylabel('Spending Score (Scaled)')

ax2.legend()


plt.tight_layout()

plt.show()
```



5. Tri-Feature Clustering: Analysis of Age, Income, and Spending Behavior

```
import plotly.express as px

# We need to re-run K-means with 3 features to make the 3D plot meaningful

X_3d = df.iloc[:, [2, 3, 4]].values # Age, Annual Income, Spending Score
```

```
kmeans_3d = KMeans(n_clusters=5, init='k-means++', random_state=42)

df['Cluster'] = kmeans_3d.fit_predict(X_3d)


# Creating an interactive 3D Plot

fig = px.scatter_3d(df, x='Age', y='Annual Income (k$)', z='Spending Score (1-100)',

        color='Cluster',

        title='3D Customer Segmentation',

        labels={'Cluster': 'Segment ID'},

        opacity=0.8,

        color_continuous_scale='Viridis')

fig.show()
```



CONCLUSION

K-Means and Hierarchical Clustering were successfully implemented to segment mall customers. K-Means proved to be more efficient for rapid segmentation, while the Hierarchical Dendrogram provided a clear visual justification for the number of clusters chosen. The resulting segments allow for data-driven marketing strategies tailored to specific consumer groups.